

Instructions for AI & optimization: project

Isep Paris, 19th May 2026

Classification of cells on histological images

Deadline: Tuesday, June 2nd 11:30 am

Work alone, or in teams of 2 or 3 students

I. Goal

The goal of this project is to develop a Machine Learning-based approach to automatically classify patches extracted from histological images of same resolution. The grading will particularly take into account the attention made on the optimization process and hyperparameter setting.

II. Training phase

The project is divided into two steps: training and test.

The training phase takes place during the 9th and 10th lab sessions.

The training set is made of patches of histological images, available in the « training dataset - images» folder, associated with labels, available in the « train.csv » file.

Your work during the training phase may be divided into up to four parts:

a) Data exploration (**mandatory**)

Get familiar with the dataset. You can for example create a « quality control » folder, where the images are split into subfolders according to their associated labels. It helps you visualizing the difficulty of the task, and choosing relevant features. You can also choose to discard some images that look irrelevant for training... but be careful, similar examples could appear in the future test set!

In your notebook (see section IV), you have to write down the:

- **total number of available training samples,**
- **labels' names and distribution in the training dataset,**
- **common appearance of a patch according to the associated label.**

You can also plot any figure relevant for your analysis (scatter plots with labels, histograms, etc.).

b) Feature extractions (**mandatory**)

Extract **homemade** features from each training image. Thus, **you have to implement your own features!** You can choose features implemented in the previous labs, and use very basic libraries such as numpy.

More especially, you should include features associated with:

- **area,**
- **shape,**

- **intensity values.**

When all the features are collected for all the available images, you are strongly recommended to save them in a csv file.

- c) Training of a machine learning algorithm (**mandatory**)

Train machine learning algorithm(s) on the selected features to classify the associated images. **You should use at least two algorithms studied during the Optimization & AI course for classification.**

More especially, you have to be careful about the following points :

- **feature selection**
- **feature normalization**
- **hyperparameter tuning**
- **algorithm selection**
- **optimization algorithm**
- **cross-validation.**

- d) Test on the future dataset (not mandatory but recommended)

Prepare the code to apply the trained model on the test dataset. It has to automatically save the results in a « test.csv » file (same structure as « train.csv »).

III. Test phase

The test phase takes place during the last 30 minutes of the 11th lab session.

A folder containing test images («test dataset - images») will be made available on Moodle.

Apply your trained model on these images, and save the predicted labels in a « test.csv » file.

IV. Expected outputs

At the end of the 11th lab session, one single member of your team has to deposit two files on Moodle.

- a) **The ipynb code file**

A single ipynb file containing the code for the **training phase (feature extraction and ML training)**, and **test phase**. This code has to be structured - respecting the plan in section II - and well commented. More precisely, add in markdown cells any comment useful to justify your choices.

You have to clearly mention the names of all the members of your team in the code file.

As usually, assess that the following requirements are fulfilled:

- Run your Jupyter notebook before sending it back (kernel → Restart & Run all).
- Check that there is no error message.
- Do not write any space or special character in the name of your file
- Ensure that each line of code does not (roughly) composed of more than 80 characters

b) **The csv prediction file**

A csv file containing two columns, the first one for the test image identifiers, and the second one for the predicted labels. The structure is the same as in the train.csv file.

V. Grading

The grading will be divided into two parts:

- 15 points : justification of the choices made for feature and machine learning frameworks, compliance with the instructions, quality and clarity of the ipynb file.
- 5 points : **relative accuracy obtained for your group i on the test set** compared with the maximum score obtained among all the groups

$$10 * \text{Acc}_i / \max_{j \in \text{nb groups}} (\text{Acc}_j)$$